BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE
	TITLE: STANDARD CHEMICAL TESTING PROCEDURE FOR BIOFEREX BLOOD TONIC (200ml)

DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/012/F01**	**PAGE NO.:**	**Page 1 of 18**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Ferric Ammonium Citrate** **Label Claim:** **85mg/5ml**	**METHOD 1:** **Reagents:** a. 20% solution of Citric Acid b. Strong Ammonia Solution c. 80% Thioglycollic Acid d. Concentrated Hydrochloric Acid ***Standard solution**:* Weigh accurately 80mg of Ferric Ammonium Citrate CRS, add 10ml of concentrated hydrochloric acid and dilute to 100ml with water. Further dilute 5ml to 100ml with water.

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Sample solution:

To appropriate quantity of the sample, add 10ml of concentrated hydrochloric acid and then dilute with water.

Procedure:

To 10ml each of the sample and standard solutions, add 2ml of Citric Acid solution, 0.1ml of Thioglycollic acid and 5ml of strong Ammonia solution, dilute to 25ml with water.

Measure the Absorbance of both solutions at 525nm against reagent blank. This will give Total Quantity (mg) of Iron in the sample.

% Assay of Ferric Ammonium Citrate = $\frac{absorbance of sample}{absorbance of standard}$ x 100

Note: 80 mg of Ferric Ammonium Citrate contains 17.2mg of elemental Iron.

METHOD 2

Digest accurately weighed quantity of the sample with 4-5ml of concentrated hydrochloric acid. Transfer to 250ml flask with the help of water. Add 15ml of

		Potassium Iodide solution (10%w/v). Keep in the dark for 5 min and titrate the liberated iodine with 0.1N Sodium thiosulphate using starch paste as indicator, added near the end titration. Each ml of 0.1N Sodium thiosulphate is equivalent to 5.585mg of Iron. Calculate the contents of FAC in the sample considering each 80mg of FAC equivalent to 17.2mg of Iron. $\% \boldsymbol{Assay}$ $= \frac{(Average\ Titre)\ x\ Standard\ Factor\ x \quad Equivalent\ weight}{85}\ x\ 100$ ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
3.	**Folic Acid** **Label Claim:** **250mcg/5ml**	**METHOD 1:** This method is based on the reaction of Folic acid with formaldehyde to yield a coloured **Reagents:** a. 0.01N Sodium Hydroxide

b. 1M Sulphuric Acid

Standard solution:

100mcg/ml of Folic Acid in 0.01N Sodium Hydroxide.

Sample solution:

To appropriate quantity of the sample equivalent to 5mg of Folic acid and extract with 25ml of 0.01N Sodium Hydroxide by shaking for 15 minutes, filter and dilute to 50ml with water.

Procedure**:**

To 10ml each of the sample and standard solutions, add 13ml of 1M Sulphuric acid and 3ml of formaldehyde solution. Close the tube with rubber stopper in which a hypodermic needle has been inserted. Heat on a boiling water – bath for 1 hour, cool and dilute to 25ml with water. Measure the absorbance of sample and standard at maximum at about 360nm against a reagent blank prepared by adding 13ml of 1M Sulphuric acid to 10ml of sample solution and diluting to 25ml with

		water. **% Assay of Folic Acid** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
4.	**Vitamin B_1** **Label Claim:** **2mg/5ml**	**METHOD 1: Reagents:** a. 0.04%w/v solution of p-amoniphenol in ethyl alcohol. b. 0.1M Ammonium Hydroxide solution ***Standard solution:*** Dissolve 100mg of Thiamine Hydrochloride (previously dried at 100°C) in distilled water to produce 100mL. Dilute this solution suitably with water to give a concentration of 20mcg/mL ***Sample solution:*** Dissolve and dilute suitably with water to give a concentration of 20mcg/mL

Procedure:

To 3ml each of the sample and standard solutions, add 3ml of 0.1N Ammonium Hydroxide solution and 0.5ml of p-aminophenol reagent. Wait for 5 minutes and extract the coloured complex with 10ml of Chloroform. Measure the Absorption of Organic layer at about 430nm against reagent blank. The contents of thiamine are calculated by comparison. The Method obeys Beer's law in the concentration range of 3 – 12 mcg/ml in final solution

The following compounds do not interfere in the above method. The figures in brackets indicate the number of folds excess the compounds can be tolerated.

- Pyridoxine
- Niacinamide
- Riboflavin
- Vitamin B12
- Ascorbic Acid

		% Assay of Vitamin B_1 = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
5.	**Vitamin B_2** **Label Claim:** **2mg/5ml**	**METHOD 1:** **Reagents:** a. 0.1N Sodium Hydroxide b. Phosphate Buffer (1 M), pH 6.0, c. Glacial Acetic Acid d. 2% w/v solution of Potassium Permanganate e. Hydrogen Peroxide (30%) ***Standard solution:*** 150mcg/mL of Riboflavin phosphate CRS in 0.1N HCl ***Sample solution:*** Accurately weigh a quantity of the sample equivalent to 15mg of the compound and extract with acid to make 100mL

		Procedure: To 3ml each of the sample and standard solutions, add 1 drop of Phenolphthalien and neutralize with 0.1N sodium hydroxide. Add 10mL of buffer solution followed by 2mL of Acetic Acid, 1mL of Potassium Permanganate solution. Allow to stand for 2 minutes, neutralize pink colour with 1mL of Hydrogen Peroxide. Shake vigorously and dilute to 25mL with water. Measure the Absorption of both the solutions at about 445nm and calculate the results by comparison **% Assay of Vitamin B_2** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 **Note:** The method is application to most of the Vitamin combinations and gives good results. ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
6.	**Vitamin B_6** **Label Claim:**	**METHOD 1: Reagents:** a. 0.05% solution of p-nitroaniline: Dissolve 0.05g of p-nitroaniline in

	2mg/5ml	5ml of methanol and dilute to 100ml with 0.033M orthophosphoric acid. b. 0.1% w/v solution of Sodium Nitrite in Water c. Phosphate Buffer, Ph 8.0: Adjust the pH of 0.5M solution of Disodium Hydrogen Phosphate to 8.0 with Sodium Hydroxide or orthophosphoric acid. Acetate Buffer, pH 4.25: Prepare 5% solution of sodium acetate and adust the pH to 4.25 with acetic acid. Mercuric Acetate solution: Dissolve 0.1589 of Mercury (iii) acetate in actetate buffer of pH 4.25 0.2% solution of Sodium carbonate in Water. ***Standard solution:*** 10mcg/mL of Pyridoxine Hydrochloride in water. ***Sample solution:*** Weigh accurately equivalent of 10mg of the substance and extract with water to produce 100ml. Further dilutions are done with water to get solution of 10mcg/ml.

For oily preparations, suspent the sample in water and extract with petroleum ether (40-60oC) and use aqueous layer for analysis.

Procedure:

Take 10ml of the sample and standard solutions, add 2.5ml of phosphate buffer (8.0). In separate test tube mix 20ml of p-nitroaniline solution and 20ml of sodium nitrite solution. Add 2.5mlof this solution to 10ml each of sample and standard solutions followed by 5ml of Mercury (ii) acetate solution. Make up the volume to 25ml and measure the absorbance at 530nm against blank.

Explanatory Note

Diazotized p-nitroaniline couples with pyridoxine at position at position para to phenolic hydroxide in phosphate buffer. The complex in turn reacts with Hg++ in acetate buffer to give azo dye (λmax 530nm).

Note:

Since the colour of the azo dye was stable for only 10 min, the colour was stabilized by complexing with Hg++. The developed colour was stable for 24hr.

		Further formation of Hg++ complex resulted in bathochromic (490 to 530nm) shift and hyperchromic effect (2-fold). Vitamin B1, B2, nicotinic acid, folic acid, Vitamin C and fat-soluble Vitamins did not interfere. **% Assay of Vitamin B_6** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
7.	**Nicotinamde** **Label Claim:** **5mg/5ml**	**METHOD 1:** **Reagents:** a. Sulphanilic acid buffered solution: To 2.5g sulphanilic acid, add 15ml water and 2ml dilute ammonia solution. Mix, add with stirring more dilute ammonia solution if necessary to dissolve the substance, adjust pH of the solution to 4.5 with dilute ammonia b. 10% w/v solution of cyanogens bromide in water. c. Dilute ammonia solution: Dilute 2ml strong ammonia solution to

100ml with water.

Standard solution:

100mcg/ml of Nicotinamide in water.

Sample solution:

Weigh equivalent to 10mg of the substance, suspend in water to make 100ml, stir and filter.

Procedure:

To 2ml each of sample and standard solutions, add 2ml of sulphanilic acid buffered solution, 5ml water, shake well and add 2ml of cyanogens bromide solution. Measure the entinction of both sample and standard solutions at 450nm after 2min against reagent blank and calculate the contents of Nicotinamide by comparison.

% Assay of Nicotinamide = $\frac{absorbance of sample}{absorbance of standard}$ x 100

REFERENCE:

Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4

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8.	**Copper Sulphate** **Label Claim:** **0.4mg/5ml**	**METHOD 1:** Assay Accurately weigh quantity of the sample equivalent to 1.25mg of copper sulphate is sulphate is ignited at 450oC, add 5ml of hydrochloric acid and evaporate to dryness, add 2.5ml of nitric acid and evaporate. Extract the residue with three 5ml portions of hot hydrochloric acid, combine the acid extract and dilute to 25ml with water. To 1ml of this solution add 10ml of edentate – citrate solution, 0.1ml of thymol blue solution and sufficient dilute ammonia solution to produce green to blue-black colour. Add 10ml sodium diethyldithiocarbamate solution and extract the yellow colour with four 5ml portions of carbon tetrachloride. Measure the absorption at maxima at about 435nm. Use standard solution of copper sulphate for comparison. **% Assay of Copper Sulphate** = $\frac{absorbance of sample}{absorbance of standard}$ x 100
9.	**Zinc Sulphate** **Label Claim:**	**METHOD 1:** **Reagents:**

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	2mg/5ml	a. Zincon Reagent: Dissolve 60mg of Zincon in 2ml of 0.2N Sodium hydroxide and 10ml distilled water. Dilute to 50ml with water, stable only for 2 days at room temperature. b. Buffer solution, pH 9.2: Dissolve 12.4g of boric acid and 15g of potassium chloride in 86ml of 1N sodium hydroxide. Dilute to 1000ml with water. ***Standard solution***: 40 mcg/ml of zinc sulphate in water. ***Procedure*** The accurately measured volume of sample containing 200mcg of Zinc sulphate and 5ml of standard solution, add 50ml of buffer solution, 2ml of zincon reagent and dilute to 100ml with buffer. Mix and measure the absorbance of sample and standard solutions at about 625nm against reagent blank and compare the results. **% Assay of Zinc Sulphate** = $\frac{absorbance of sample}{absorbance of standard}$ x 100

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		REFERENCE: Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
10.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an "S" or "8" shape. - Allow the media to completely solidify. - Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of

colonies, it is required to carry out identification test.

- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\textbf{CFU/mL} = \frac{\textbf{Total number of colonies obtained x dilution factor}}{\textbf{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats.
- The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105**
- If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

*LAF--- Laminar Air Flow

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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